

DESIGN OF LOW-POWER HIGH-PERFORMANCE 2–4 AND 4–16 MIXED-LOGIC LINE DECODERS

Abstract:

This brief introduces a mixed-logic design method for line decoders, combining transmission gate logic, pass transistor dual-value logic, and static complementary metal-oxide semiconductor (CMOS). Two novel topologies are presented for the 2–4 decoder: a 14-transistor topology aiming on minimizing transistor count and power dissipation and a 15-transistor topology aiming on high power-delay performance. Both normal and inverting decoders are implemented in each case, yielding a total of four new designs. Furthermore, four new 4–16 decoders are designed by using mixed-logic 2–4 predecoders combined with standard CMOS postdecoder. All proposed decoders have full-swinging capability and reduced transistor count compared to their conventional CMOS counterparts. Finally, a variety of comparative spice simulations at 32 nm show that the proposed circuits present a significant improvement in power and delay, outperforming CMOS in almost all cases.

TOOLS:

➤ Xilinx ISE 14.7

LANGUAGE:

➤ Verilog HDL

TEMPOSTOP time flow shower kit

Time flow, anti-blocking system (AB)

Cross-wall 200 mm, AB, ROUND shower head

Reference: **747110**

Specification description

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Time flow shower kit for cross wall installation:

TEMPOSTOP time flow valve M $\frac{1}{2}$ " for mixed water supply.

For walls $\leq$ 200mm.

Time flow $\sim$ 15 seconds.

Flow rate 6 lpm at 3 bar.

Chrome-plated ROUND shower head, tamperproof and scale-resistant with automatic flow rate regulation.

Chrome-plated metal control.

Concealed fixing and anti-rotating pin.

Threaded rods can be cut to size.

AB anti-blocking system: water only flows when push button is released.

30-year warranty.

[Formerly ref: 747214]



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